

9.5.2.

Use theorem 9.5.1 to compute each of the following.

a. $\binom{6}{0}$ b. $\binom{6}{1}$ c. $\binom{6}{2}$

d. $\binom{6}{3}$ e. $\binom{6}{4}$ f. $\binom{6}{5}$ g. $\binom{6}{6}$

a. $\binom{6}{0}$

$$\binom{6}{0} = \frac{6!}{0!(6-0)!} = \frac{6!}{0! \times 6!} = \frac{6!}{1 \times 6!} = 1$$

Solution: a. $\binom{6}{0} = 1$

b. $\binom{6}{1}$

$$\binom{6}{1} = \frac{6!}{1!(6-1)!} = \frac{6!}{1! \times 5!} = \frac{6!}{5!} = \frac{6 \times 5!}{5!} = \frac{6}{1} = 6$$

Solution: b. $\binom{6}{1} = 6$

c. $\binom{6}{2}$

$$\binom{6}{2} = \frac{6!}{2!(6-2)!} = \frac{6!}{2! \times 4!} = \frac{6 \times 5 \times 4!}{2! \times 4!} = \frac{6 \times 5}{2!} = \frac{30}{2} = 15$$

Solution: c. $\binom{6}{2} = 15$

d. $\binom{6}{3}$

$$\binom{6}{3} = \frac{6!}{3!(6-3)!} = \frac{6!}{3! \times 3!} = \frac{6 \times 5 \times 4 \times 3!}{3! \times 3!} = \frac{6 \times 5 \times 4}{3!} = \frac{120}{3 \times 2!} = \frac{120}{6} = 20$$

Solution: d. $\binom{6}{3} = 20$

e. $\binom{6}{4}$

$$\binom{6}{4} = \frac{6!}{4!(6-4)!} = \frac{6!}{4! \times 2!} = \frac{6 \times 5 \times 4!}{4! \times 2!} = \frac{6 \times 5}{2!} = \frac{30}{2} = 15$$

Solution: e. $\binom{6}{4} = 15$

f. $\binom{6}{5}$

$$\binom{6}{5} = \frac{6!}{5!(6-5)!} = \frac{6!}{5! \times 1!} = \frac{6!}{5!} = \frac{6 \times 5!}{5!} = \frac{6}{1} = 6$$

Solution: f. $\binom{6}{5} = 6$

g. $\binom{6}{6}$

$$\binom{6}{6} = \frac{6!}{6!(6-6)!} = \frac{6!}{6! \times 0!} = \frac{6!}{6!} = 1$$

Solution: g. $\binom{6}{6} = 1$